



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

AIOT

26-03-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	B	C	B	D	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	A	A	D	C	A	B	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	8	34	10	5	8	25	800	12	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	B	D	C	D	B	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	D	B	B	B	D	B	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	10	14	600	4	3	9	7	7	4	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	C	C	A	B	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	C	B	D	C	C	A	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1274	72	5	90	50	2	99	18	10	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

It is a case of resonance,

$$\therefore X_L = X_C$$

$$\Rightarrow Z = R$$

$$\therefore I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200}{100} = 2\text{A}$$

$$\therefore P_{\text{av}} = I_{\text{rms}}^2 R = 4 \times 100 = 400\text{W}$$

2. Ans (B)

The total charge

$$Q = \int_0^{2\pi} \lambda \cos^2 \theta R d\theta = \pi \lambda R$$

$$\text{Hence, } \pi \lambda_1 R_1 = \pi \lambda_2 R_2$$

$$\frac{\lambda_1}{R_2} = \frac{\lambda_2}{R_1}$$

3. Ans (C)

Radius of the orbit

$$r = (0.529) \frac{n^2}{Z} \text{Å} = \frac{4^2}{3} (0.529) \text{Å}$$

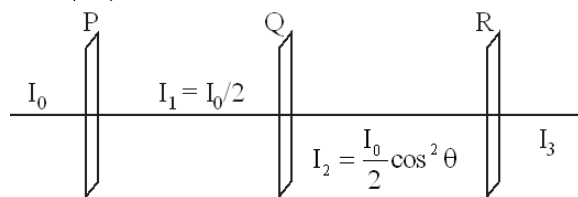
Circumference of orbit,

$$c = 2\pi r = 1.77 \times 10^{-9} \text{m}$$

$$\text{Electron's speed} = (2.18 \times 10^6) \frac{3}{4}$$

$$\therefore N = \frac{10^{-8}}{(c/v)} = 9.2 \times 10^6 \text{ revolution}$$

4. **Ans (B)**



$$I_1 = \frac{I_0}{2}$$

$$I_2 = \frac{I_0}{2} \cos^2 \theta$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \cos^2 (90 - \theta)$$

$$\theta = 0 \rightarrow I_3 = 0$$

$$\theta = 45^\circ \rightarrow I_3 = \frac{I_0}{2} \cos^2 45^\circ \cdot \cos^2 45^\circ$$

$$\frac{I_0}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{I_0}{8}$$

$$\theta = 90^\circ \rightarrow I_3 = 0$$

5. **Ans (C)**

$$r_1 = \frac{mv \sin 53}{qB} = \frac{m \times 8}{qB} \times \frac{4}{5} = \frac{32m}{5qB}$$

$$r_2 = \frac{m \times 18}{5qB},$$

$$\text{maximum separation} = 2(r_1 + r_2) = \frac{20m}{qB}$$

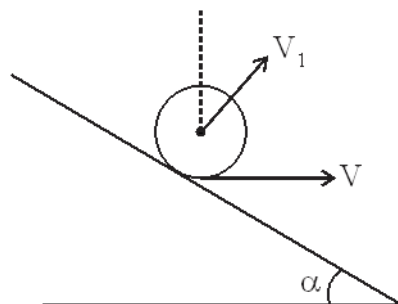
Both have same pitch hence meet after one revolution.

$$d = 8 \times \frac{3}{5} \times \frac{2\pi m}{Bq} = \frac{9.6 \pi m}{Bq}$$

6. **Ans (B)**

After removal of one electron, remaining atom is hydrogen like atom. Its energy in first orbit is $E = -2^2 (13.6) = -54.4 \text{ eV}$. So, binding energy = $79 - 54.4 = 24.6 \text{ eV}$.

7. **Ans (D)**



$$e = \frac{V_1 - V \sin \alpha}{V \sin \alpha} = 1$$

$$V_1 = 2V \sin \alpha$$

8. **Ans (C)**

We know that

$$P_B = P_A + \frac{1}{2} \rho \omega^2 a^2$$

$$P_D = P_A + \rho g a$$

$$P_C = P_D + \frac{1}{2} \rho \omega^2 a^2 = P_A + \rho g a + \frac{1}{2} \rho \omega^2 a^2$$

Therefore,

$$P_C > P_A \text{ for all values of } \omega \text{ and } P_B > P_D \text{ only}$$

$$\text{if } \omega > \sqrt{\frac{2g}{a}}$$

9. **Ans (C)**

$$T = T_0 + \alpha V$$

$$\therefore \frac{dT}{dV} = \alpha$$

$$\therefore C = \frac{3R}{2} + \frac{R(T_0 + \alpha V)}{V} \left(\frac{1}{\alpha} \right)$$

$$= \frac{3R}{2} + \frac{RT_0}{\alpha V} + R = \frac{5R}{2} + \frac{RT_0}{\alpha V}$$

$$\therefore \Delta Q = (1) \int \left(\frac{5R}{2} + \frac{RT_0}{\alpha V} \right) dT$$

$$= \alpha \int_{V_0}^{2V_0} \left(\frac{5R}{2} + \frac{RT_0}{\alpha V} \right) dV$$

$$= \frac{5R\alpha V_0}{2} + RT_0 \ln(2)$$

10. **Ans (C)**

$$K_i + U_i = 0$$

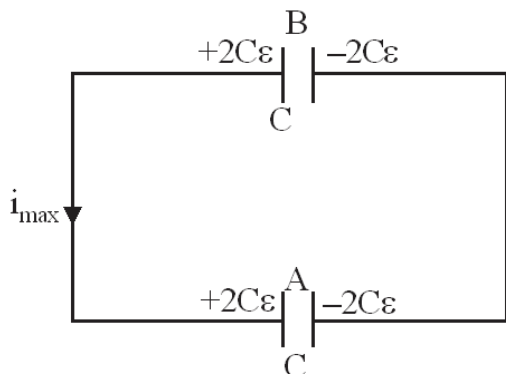
$$\frac{1}{2} m v_e^2 - \frac{GMm}{R} \left(\frac{3}{2} - \frac{r^2}{2R^2} \right) = 0$$

11. **Ans (A)**

When the current through the inductor is maximum, the two capacitors will be in parallel.

Now,

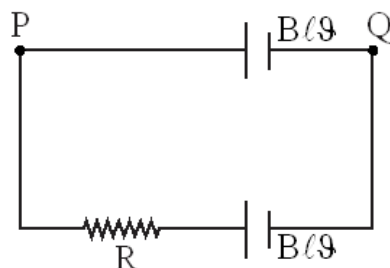
$$\frac{(3C\varepsilon)^2}{2C} + \frac{(C\varepsilon)^2}{2C} = 2 \times \frac{(2C\varepsilon)^2}{2C} + \frac{1}{2} Li_{\max}^2$$



$$\Rightarrow \frac{9C\varepsilon^2}{2} + \frac{C\varepsilon^2}{2} = 4C\varepsilon^2 + \frac{1}{2} Li_{\max}^2$$

$$\Rightarrow i_{\max} = \varepsilon \sqrt{\frac{2C}{L}}$$

12. **Ans (C)**



$$V_{PQ} = B\varepsilon$$

$$\therefore \text{net emf} = 0$$

$$\therefore I = 0$$

13. **Ans (A)**

$$\vec{E} = (2\hat{k} - 3\hat{j}) \times 10^{-3} \sin [10^7 (x + 2y + 3z - \beta t)]$$

$$\vec{k} = 10^7 (\hat{x} + 2\hat{y} + 3\hat{z}) \Rightarrow |\vec{k}| = 10^7 \sqrt{14},$$

$$\omega = 10^7 \beta,$$

$$c = \frac{\omega}{|\vec{k}|} = \frac{10^7 \beta}{10^7 \sqrt{14}} \Rightarrow \beta = \sqrt{14}c$$

14. **Ans (A)**

$$V_1 \rho_1 g + V_2 \rho_2 g = (V_1 + V_2) \frac{3}{2} \rho g$$

$$V_1 : V_2 = 1 : 1$$

15. **Ans (D)**

$$r = \sqrt{1+1+4} = \sqrt{6} < 5$$

$$\Rightarrow E_{\text{in}} = 0$$

16. **Ans (C)**

$$t = 0, N = N_0,$$

$$t = 6.93, N = N_0/4$$

$N_0/4$ is the sample left after two half-lives

$$\therefore 2t_{1/2} = 6.93$$

$$\Rightarrow 2 \times \frac{0.693}{\lambda} = 6.93$$

$$\Rightarrow \lambda = 0.2 \text{ min}^{-1}$$

$$\Rightarrow t = 60 \text{ min}$$

$$\therefore N = N_0 e^{-\lambda t}$$

$$= N_0 e^{-0.2 \times 60} = \frac{N_0}{e^{12}}$$

17. **Ans (A)**

Here, $R_{\text{in}} = 1000 \Omega$,

$$\Delta I_B = 10 \mu\text{A} = 10^{-5} \text{ A},$$

$$\Delta I_C = 2 \text{ mA} = 2 \times 10^{-3} \text{ A},$$

$$R_{\text{out}} = 5 \text{ k}\Omega = 5 \times 10^3 \Omega$$

(a) Current gain,

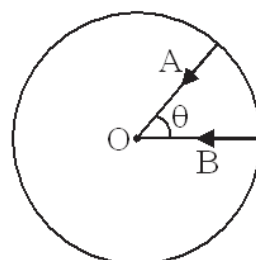
$$\beta = \frac{\Delta I_C}{\Delta I_B} = \frac{2 \times 10^{-3} \text{ A}}{10^{-5} \text{ A}} = 200$$

(b) Voltage gain,

$$A_v = \beta \left(\frac{R_{\text{out}}}{R_{\text{in}}} \right) = 200 \left(\frac{5 \times 10^3 \Omega}{1000 \Omega} \right) = 1000$$

18. **Ans (B)**

$$f_A = f_s \left(\frac{v_s}{v_s - v_A} \right),$$

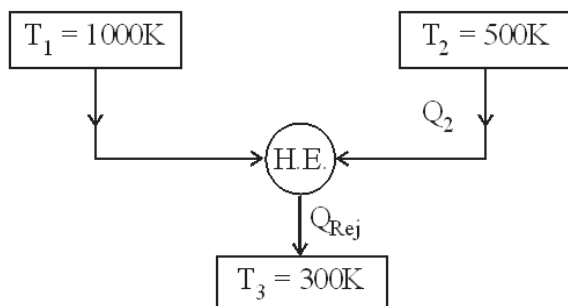


$$f_B = f_s \left(\frac{v_s}{v_s - v_B} \right)$$

$$v_A = v_B < v_s$$

$$\therefore f_A = f_B > f_s$$

19. Ans (A)



From Clausius inequality for reversible heat engine,

$$\Delta S_{\text{rev}} = 0$$

$$\frac{Q}{1000} + \frac{Q}{500} - \frac{Q_R}{300} = 0$$

$$\frac{3Q}{1000} = \frac{Q_R}{300}$$

$$\frac{Q_R}{Q} = \frac{9}{10}$$

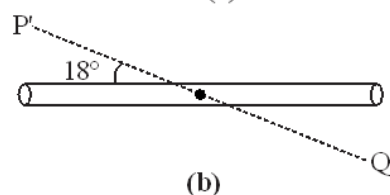
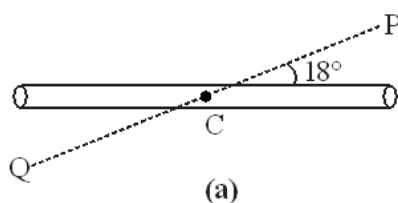
Divided by 2 in both side

$$\frac{Q_R}{2Q} = \frac{9}{20}$$

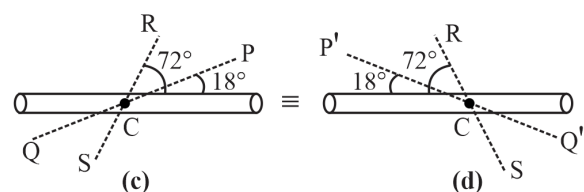
$$1 - \frac{Q_R}{2Q} = 1 - \frac{9}{20} = \frac{11}{20}$$

$$\eta_{\text{HE}} = 55\%$$

20. Ans (D)



The MI of rod about axis PQ [Figure (a)] and MI of rod about axis P'Q' [Figure (b)] are the same by symmetry.



$$\therefore I_{PQ} + I_{RS} = I_{P'Q'} + I_{RS} = \frac{ml^2}{12}$$

by perpendicular axis theorem.

PART-1 : PHYSICS

SECTION-II

1. Ans (6)

$$x = +\frac{A}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\Delta\theta = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

$$t = \frac{\Delta\theta}{\omega} = \frac{T}{6}$$

2. Ans (8)

Initial speed is the maximum speed

$$\Rightarrow u = 20 \text{ m/s}$$

Speed at highest point is minimum

$$\Rightarrow u \cos\theta = 10 \Rightarrow 20 \cos\theta = 10 \Rightarrow \theta = 60^\circ$$

Radius of curvature is minimum at highest point.

$$a_N = \frac{v^2}{R} \Rightarrow R_{\min} = \frac{(u \cos\theta)^2}{g} = \frac{10^2}{10} = 10$$

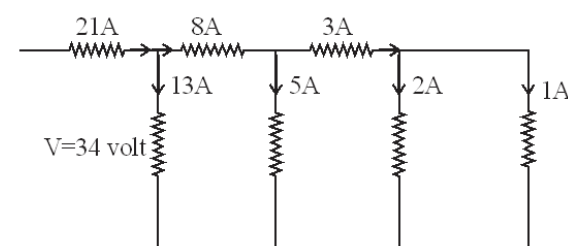
R is maximum at the initial point.

$$a_N = \frac{v^2}{R} \Rightarrow g \cos\theta = \frac{u^2}{R_{\max}}$$

$$\Rightarrow R_{\max} = \frac{u^2}{g \cos\theta} = \frac{20^2}{10 \times \frac{1}{2}} = 80$$

$$\frac{R_{\max}}{R_{\min}} = \frac{80}{10} = 8$$

3. Ans (34)



4. Ans (10)

According to the problem, power = 1W.

Zener breakdown voltage, $V_z = 5V$

Minimum voltage, $V_{\min} = 3V$

Maximum voltage, $V_{\max} = 7V$

We know, $P = VI$

$$\text{So, current } I_{Z_{\max}} = \frac{P}{V_z} = \frac{1}{5} = 0.2A$$

For safe operation,

$$R_s = \frac{V_{\max} - V_z}{I_{Z_{\max}}} = \frac{7 - 5}{0.2} = 10 \Omega$$

5. **Ans (5)**

$$I = \frac{24}{\frac{RG}{R+G} + 1}$$

$$V = 24 - \text{potential difference across } 1 \Omega$$

$$= 24 - I$$

$$V = 24 - \left(\frac{24}{\frac{1}{\frac{1}{R} + \frac{1}{G}} + 1} \right)$$

$$\text{For } G \rightarrow \infty \Rightarrow \frac{1}{G} \rightarrow 0 \text{ and } V = 20 \text{ volts}$$

$$\Rightarrow 20 = 24 - \frac{24}{R+1} \Rightarrow R = 5 \Omega$$

6. **Ans (8)**

Since there is no external force in horizontal direction $\vec{P}_i = \vec{P}_f$

P_i = Momentum of monkey in horizontal direction just before reaching heap of fruits for first time.

P_f = Momentum of monkey and fruits in horizontal direction just after passing over heap for fifth time.

$$P_i = M\sqrt{2gl} = (20\text{kg})\sqrt{2 \times 10\text{m/s}^2 \times 5\text{m}}$$

$$= 200\text{kgm/s}$$

$$P_f = (M + m)v = (20\text{kg} + 5\text{kg})v$$

$$200 = 25v \Rightarrow v = 8\text{m/s}$$

7. **Ans (25)**

Moment of inertia of the cylinder about the central axis,

$$I = \frac{1}{2}mR^2 = \frac{1}{2}(2)(0.2)^2 = 0.04 \text{ kg} - \text{m}^2$$

$$\text{Now, } \tau = I\alpha = I \frac{d\omega}{dt}$$

$$\Rightarrow \omega = \frac{1}{I} \int_0^{10} \tau dt = \frac{1}{I} \text{ (Area under torque - time graph)}$$

$$= \frac{1}{0.04} \left(\frac{1}{2} (0.2) (10) \right) = 25 \text{ rad/s}$$

8. **Ans (800)**

Let the mass of ice needed be x

We know that,

Heat absorbed by ice to rise to 0°C

= (Heat lost by calorimeter to fall to 0°C + Heat lost by water to fall to 0°C and freeze)

$$\text{Therefore, } x(0.5)(0 - (-40)) = 50(1)(20 - 0) + 150(1)(20 - 0) + 150(80)$$

Solving, we get $x = 800 \text{ g}$

10. **Ans (9)**

$$\text{Heat radiated} = \epsilon\sigma AT^4 = KT^4$$

$$-mS(dT/dt) = KT^4$$

$$-mS \int_{200}^{100} \frac{dT}{T^4} = Kt_1$$

$$t_1 = \frac{1}{K_1} \left[\frac{1}{100^3} - \frac{1}{200^3} \right] = \frac{1}{K_1} \left[\frac{7}{200^3} \right]$$

$$t_2 = \frac{1}{K_1} \left[\frac{1}{50^3} - \frac{1}{200^3} \right] = \frac{1}{K_1} \left[\frac{63}{200^3} \right]$$

$$\frac{t_2}{t_1} = 9$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (C)**

Unit cell formula before modification

$\text{C}^{2-} \rightarrow 3$ per unit cell

$\text{A}^{2+} \rightarrow 1$ per unit cell

$\text{B}^+ \rightarrow 4$ required per unit cell to maintain charge neutrality, i.e., alternate tetrahedral voids are occupied.

If all ions along a body diagonal are removed, then ions removed :

$\text{C}^{2-} \rightarrow \text{zero}$

$$\text{A}^{2+} \rightarrow 2 \times \frac{1}{8} = \frac{1}{4}$$

$\text{B}^+ \rightarrow 1$

Effective no of ions remaining in the unit cell

$\text{C}^{2-} \rightarrow 3$

$$\text{A}^{2+} \rightarrow 1 - \frac{1}{4} = \frac{3}{4}$$

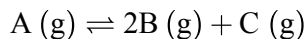
$\text{B}^+ \rightarrow 4 - 1 = 3$

Resulting formula $\text{A}_{3/4}\text{B}_3\text{C}_3$ or AB_4C_4 .

2. **Ans (A)**

The above equation is virial equation of state for real gas. The value of $A = RT$ at any given temperature for any gas.

3. **Ans (B)**



$$t = 0 \quad 1 \quad - \quad -$$

$$t = t_{eq} \quad 1 - \alpha \quad 2\alpha \quad \alpha$$

$$D_i = \frac{M}{2} (m = \text{mol. unit of } A)$$

$$D_f = \frac{M}{2(1 + 2x)}$$

$$\Rightarrow \frac{D_i}{D_f} = 1 + 2x$$

$$\Rightarrow x = \frac{D_i}{2D_f} - \frac{1}{2}$$

4. **Ans (B)**

Radial probability distribution function $P(r)$ gives probability of finding electron anywhere between 2 walls of spherical shell of thickness dr at the radius r .

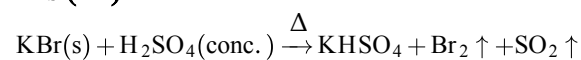
6. **Ans (C)**

Cu containing low grade ore can not be leached out by alkali, its is done by acid.

7. **Ans (D)**

Presence of H^- ions in saline hydrides is confirmed by liberation of hydrogen gas at anode on electrolysis of their melts.

8. **Ans (B)**



9. **Ans (A)**

Clean water would have BOD value of less than 5 ppm where as highly polluted water have BOD value of 17 ppm or more.

10. **Ans (C)**

In lanthanoid series, chemical behaviour of earlier members is similar to calcium but latter members behave more like aluminium.

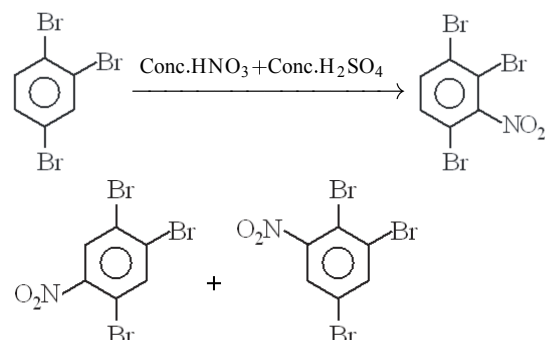
11. **Ans (A)**

Due to stable configuration metallic radius of Mn is higher.

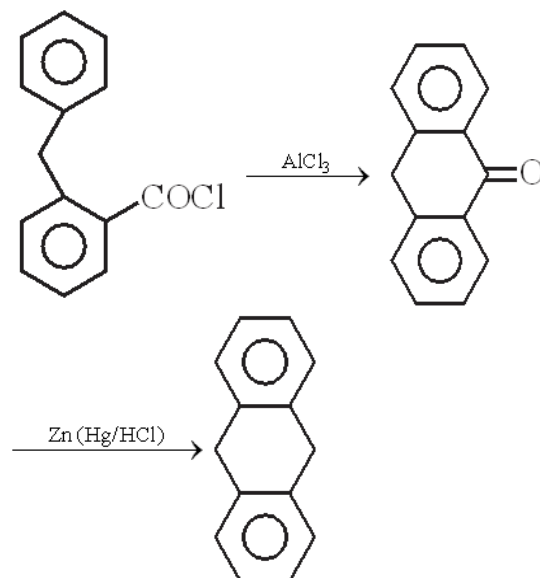
12. **Ans (D)**

In complex P all donor atoms of ligands are not at equal distance from central metal.

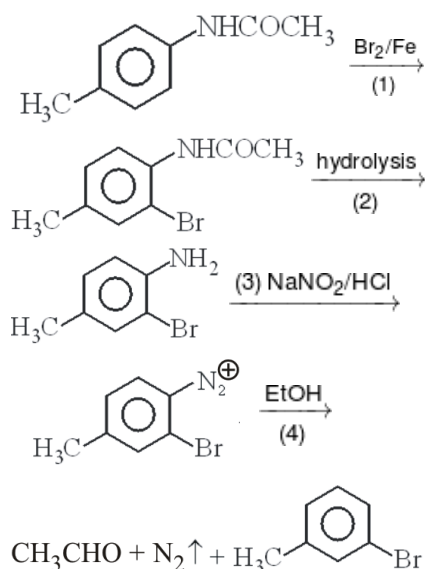
14. **Ans (B)**



15. **Ans (B)**



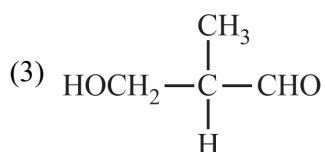
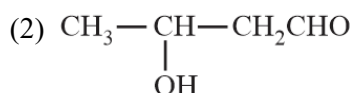
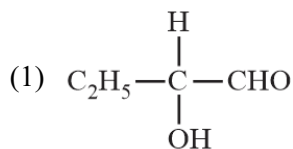
17. **Ans (D)**



19. **Ans (B)**

Rate $\propto$ [Stability of carbocation]

20. **Ans (C)**



PART-2 : CHEMISTRY

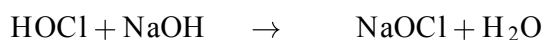
SECTION-II

1. **Ans (10)**

After addition of 0.2M NaOH, since $\text{pH} = \text{pK}_a$, a buffer consisting of equal moles of HOCl and NaOCl has been formed

$$\left(\text{pH} = \text{pK}_a + \frac{[\text{NaOCl}]}{[\text{HOCl}]} \right)$$

Let volume of original HOCl solution = V ml



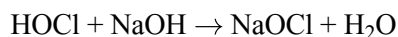
in mmols 0.2V (20 × 0.2) = 4

Eq. (0.2V - 4) - 4

Now, 0.2V - 4 = 4

$\Rightarrow V = 40\text{ml}$

After addition of 40 ml NaOH



mmol 0.2 × 40 0.2 × 40

Eq - - 8

$$\therefore [\text{NaOCl}] = \frac{8}{40 + 40} = 0.1 \text{ M}$$

$$\text{pH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$$

$$= \frac{1}{2}(14 + 7 - 1)$$

$$= 10$$

2. **Ans (14)**

Adiabatic free expansion is an isothermal process

$$\Delta S_1 = \Delta S_{\text{adiabatic free expansion}} = nR \ln \left(\frac{V_2}{V_1} \right)$$

$$= 10 \times 2 \ln \left(\frac{2V}{V} \right)$$

$$= 14 \text{ cal K}^{-1}$$

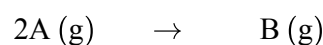
$$\Delta S_2 = \Delta S_{\text{reversible adiabatic}} = 0$$

$$\Delta S_{\text{cycle}} = \Delta S_1 + \Delta S_2 + \Delta S_3 = 0$$

$$\Rightarrow \Delta S_3 = -\Delta S_1$$

$$\therefore \Delta S_{\text{isochoric process}} = \Delta S_3 = -14 \text{ cal K}^{-1}$$

3. **Ans (600)**



$$t = 0 \quad a_0 \quad -$$

$$t = 10 \text{ min} \quad a_t = a_0 - x \quad x/2$$

$$t = t_{\infty} \quad - \quad a_0/2$$

At constant V and T, $P \propto$ no. of moles

$$\text{At } t = 0 \text{ min} \Rightarrow a_0 - x + \frac{x}{2} \propto 3$$

$$\Rightarrow \frac{2a_0 - x}{2} \propto 3 \quad \dots(1)$$

$$\text{After long time, } \frac{a_0}{2} \propto 2 \quad \dots(2)$$

$$\text{Dividing (1) by (2)} \quad \frac{2a_0 - x}{a_0} = \frac{3}{2}$$

$$\Rightarrow 4a_0 - 2x = 3a_0$$

$$\Rightarrow a_0 = 2x$$

$$\text{After } t = 10 \text{ min} \Rightarrow a_t = a_0 - x - x$$

Since conc. of [A] becomes half after 10 mins.

$$\therefore t_{\frac{1}{2}} = 10 \text{ min} \times 60 \text{ sec} = 600 \text{ sec.}$$

4. **Ans (4)**



5. **Ans (3)**

$$x = \text{N}_2^{2+}, \text{C}_2, \text{B}_2^+, \text{O}_2^+ = 4$$

$$y = \text{B}_2^+, \text{O}_2^+ = 2$$

6. **Ans (9)**

$$x = 6$$

$$y = 4$$

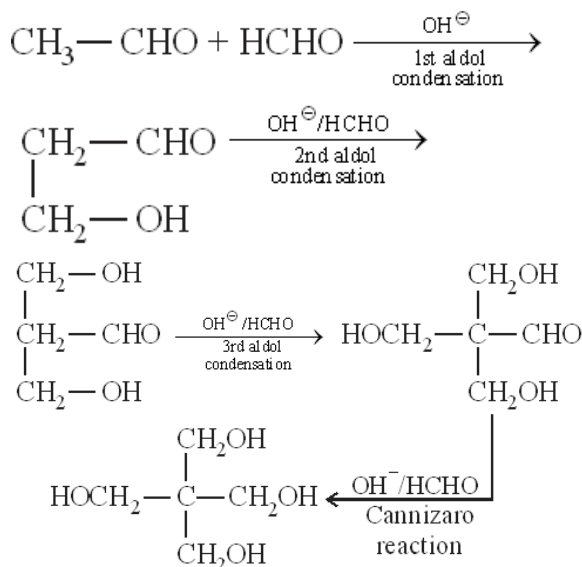
7. **Ans (7)**



8. **Ans (7)**

All except VI, IX

9. **Ans (4)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$g(f(x))$ is cubic and $g(x)$ is also cubic

$$\Rightarrow f(x) = ax + b$$

$$g(f(x)) = (ax + b)^3 + (ax + b) - 1$$

$$= a^3x^3 + 3a^2bx^2 + (3ab^2 + a)x + b^3 + b - 1$$

$$\text{Now } a^3 = 27 \Rightarrow a = 3 \text{ and } 3ab^2 + a = 39$$

$$\Rightarrow b = 2, -2$$

$$\text{if } b = 2 \quad \lambda_1 = -54 \quad (\times)$$

$$\text{if } b = -2 \quad \lambda_1 = 54 \quad (\checkmark)$$

$$\Rightarrow a = 3, b = -2$$

$$\Rightarrow \lambda_2 = -8 - 2 - 1 = -11$$

$$\Rightarrow f(x) = ax + b = 3x - 2$$

$$f(2) = 4$$

$$\Rightarrow f(2) + \lambda_1 - \lambda_2$$

$$= 4 + 54 + 11$$

$$= 69$$

2. **Ans (C)**

Points lies on same side of the line.

$$\Rightarrow (2 - 1)(\sin\theta - \cos 2\theta) > 0$$

$$\Rightarrow \sin\theta - \cos 2\theta > 0$$

$$\Rightarrow (\sin\theta - 1 + 2\sin^2\theta) > 0$$

$$\Rightarrow (2\sin^2\theta + \sin\theta - 1) > 0$$

$$\Rightarrow \sin\theta > \frac{1}{2}, \sin\theta < -1$$

$$\Rightarrow \frac{\pi}{6} < \theta < \frac{5\pi}{6}$$

3. **Ans (D)**

Let centre of the circle be $(\alpha, 1 - 2\alpha)$ and

radius = r

$$\text{Now } \frac{|\alpha - 1 + 2\alpha|}{\sqrt{2}} = r \text{ and } \frac{|\alpha + 1 - 2\alpha|}{\sqrt{2}} = r$$

$$\text{Now } |3\alpha - 1| = |\alpha - 1|$$

$$\Rightarrow a = 0, \frac{1}{2}$$

$$\Rightarrow r = \frac{1}{\sqrt{2}} (\alpha = 0) \text{ and } r = \frac{1}{2\sqrt{2}} \left(\alpha = \frac{1}{2} \right)$$

$$\Rightarrow \text{Radius of largest circle be } \frac{1}{\sqrt{2}}$$

4. **Ans (C)**

$$\begin{aligned} \sum_{r=0}^{20} {}^{50-r}C_6 + \sum_{r=1}^{25} {}^{r+5}C_6 \\ = ({}^{50}C_6 + {}^{49}C_6 + {}^{48}C_6 + \dots + {}^{31}C_6 + {}^{30}C_6) + \\ ({}^6C_6 + {}^7C_6 + {}^8C_6 + \dots + {}^{30}C_6) \\ = {}^6C_6 + {}^7C_6 + {}^8C_6 + \dots + {}^{50}C_6 + {}^{30}C_6 \\ = {}^7C_7 + {}^7C_6 + {}^8C_6 + \dots + {}^{50}C_6 + {}^{30}C_6 \\ = {}^{51}C_7 + {}^{30}C_6 \dots\dots 1 \end{aligned}$$

$$\begin{aligned} \sum_{r=0}^6 {}^{20}C_r {}^{10}C_{r+4} = \text{coefficient of } x^6 \text{ in } (1+x)^{20} \\ (1+x)^{10} = {}^{30}C_6 \dots\dots (2) \end{aligned}$$

Now from (1) & (2)

$$\begin{aligned} \sum_{r=0}^{20} {}^{50-r}C_6 + \sum_{r=1}^{25} {}^{r+5}C_6 - \sum_{r=0}^6 {}^{20}C_r {}^{10}C_{r+4} = \\ {}^{51}C_7 \text{ or } {}^{51}C_{44} \end{aligned}$$

$$\Rightarrow m + n = 58 \text{ or } 95$$

5. **Ans (C)**

$$(x-1)(x^2+1) > 0$$

$$\Rightarrow x > 1$$

$$\tan^{-1} \frac{2x}{1-x^2} = -\pi + 2\tan^{-1} x; x > 1$$

$$\sin\left(\frac{1}{2}\tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x\right) = \sin\left(-\frac{\pi}{2}\right) = -1$$

6. **Ans (A)**

$$\vec{a} = \vec{b} \times \vec{c} + 3\vec{b}$$

Taking dot product with $\vec{b}$

$$|a||b| \cos \theta = 3|b|^2$$

$$\Rightarrow \cos \theta = \frac{6}{|a|} \Rightarrow \theta = 0^\circ$$

$$\Rightarrow \vec{a} = 3\vec{b} \quad \because \vec{b} \times \vec{c} = 0$$

$$\Rightarrow \vec{b} = \vec{c} \text{ or } \vec{b} = -\vec{c}$$

$$|2\vec{a} + \vec{b} + \vec{c}| = 8|\vec{b}| \text{ or } 6|\vec{b}|$$

$$\text{Sum} = 16 + 12 = 28$$

7. **Ans (B)**

Vector parallel to plane

$$P_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i}(-1) - \hat{j}(2) + \hat{k}$$

Normal vector of plane

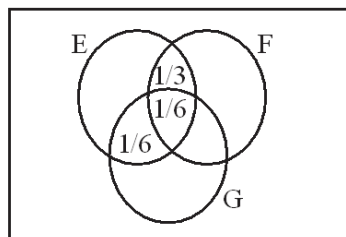
$$P_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ -1 & -2 & 1 \end{vmatrix} = \hat{i}(-3) - \hat{j}(0) + \hat{k}(-3)$$

Equation of plane

$$P_2; -3(x-1) + (z-1)(-3) = 0$$

$$x + z - 2 = 0$$

8. **Ans (C)**



$$\text{Required probability} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{2}$$

9. **Ans (B)**

$$x^3 \cdot dy + 3xy^2 \cdot dy - x \cdot e^{xy} \cdot dy$$

$$= y \cdot e^{xy} \cdot dx - 3x^2y \cdot dx - y^3 dx$$

$$\Rightarrow (x^3 dy + 3x^2y dx) + (3xy^2 \cdot dy + y^3 dx)$$

$$= x \cdot e^{xy} dy + y \cdot e^{xy} \cdot dx$$

$$\Rightarrow \int d(x^3y) + \int d(xy^3) = \int d(e^{xy})$$

$$\Rightarrow x^3y + xy^3 = e^{xy} + k$$

10. **Ans (B)**

$$y = mx - 2am - am^3$$

$$\Rightarrow 2 = 5m - 2m - m^3$$

$$\Rightarrow m^3 - 3m + 2 = 0$$

$$(m-1)^2(m+2) = 0$$

$$\Rightarrow m = 1, -2$$

$$A = \{1, -2\} \text{ \& } B = \{-1, 2\}$$

$$\Rightarrow n(A) = 2 \Rightarrow n(B) = 2$$

$$\text{No of non empty relation} = 2^{2 \times 2} - 1 = 15$$

11. **Ans (C)**

$$\sim (p \wedge (q \vee r)) = \sim ((p \wedge q) \vee (p \wedge r))$$

$$= \sim (p \wedge q) \wedge \sim (p \wedge r)$$

$$= (\sim p \vee \sim q) \wedge (\sim p \vee \sim r)$$

12. **Ans (D)**

$$\text{Median} = \frac{25a + 26a}{2} = \frac{51}{2}a$$

$$\therefore MD = 50$$

$$\frac{\sum |x_i - M|}{50} = 50$$

$$\left|a - \frac{51}{2}a\right| + \left|2a - \frac{51}{2}a\right| + \left|3a - \frac{51}{2}a\right|$$

$$+ \dots + \left|50a - \frac{51}{2}a\right| = 2500$$

$$2(1 + 3 + \dots + 49) |a| = 2500$$

$$\Rightarrow a^2 = 16$$

13. Ans (C)

$$x^3 - 3x^2 + 3x + 7 = 0 \begin{matrix} \swarrow p \\ \searrow q \\ \quad r \end{matrix}$$

$$\text{Now, } \begin{vmatrix} 2qr - p^2 & r^2 & q^2 \\ r^2 & 2pr - q^2 & p^2 \\ q^2 & p^2 & 2pq - r^2 \end{vmatrix}$$

$$= \begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} \times \begin{vmatrix} -p & r & q \\ -q & p & r \\ -r & q & p \end{vmatrix} = \begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix}^2$$

$$= -(p^3 + q^3 + r^3 - 3pqr)^2$$

$$= \{(p+q+r)[p^2+q^2+r^2-pq-qr-rp]\}^2$$

$$= \{(p+q+r)[(p+q+r)^2 - 3(pq+qr+rp)]\}^2$$

$$= \{(3)[(3)^2 - 3(3)]\}^2 = 0$$

14. Ans (B)

$$\int e^x \cdot \{e^{x^2} \cdot (x+1) + e^{x^2} \cdot 2x \cdot (x+1) + e^{x^2} \cdot 1\} dx$$

$$= e^x \cdot e^{x^2} \cdot (x+1) + c$$

15. Ans (D)

Vertices of the square lies on the director circle
 $x^2 + y^2 = 25$ & the diagonal of the square be the
 diameter of the circle.

Let λ be the side of the square

$$\Rightarrow \lambda\sqrt{2} = 2 \times 5$$

$$\Rightarrow \lambda = 5\sqrt{2}$$

$$\Rightarrow \lambda^2 = 50$$

16. Ans (C)

For rectangular hyperbola $xy = c^2$

Required area = $2c^2 = 8$

17. Ans (C)

Let α, β are the roots of the given equation and
 both α, β are prime. $\alpha + \beta = 2023$, so one of the
 root must be '2' other is 2021.

$$k = \alpha\beta = 2 \times 2021 = 4042$$

18. Ans (A)

$$g'(x) = f'(x^2 - 4) \cdot 2x - f'(2 - x^2) \cdot 2x$$

$$= 2x(f'(x^2 - 4) - f'(2 - x^2))$$

$$\Rightarrow f'(x^2 - 4) - f'(2 - x^2) > 0$$

$$\Rightarrow x^2 - 4 < 2 - x^2$$

$$\Rightarrow 2x^2 < 6$$

$$\Rightarrow x^2 < 3$$

$$\Rightarrow -\sqrt{3} < x < \sqrt{3}$$

So, $(-\sqrt{3}, 0)$

19. Ans (A)

$$T_r = \frac{1.3.5 \dots (2r-1)((2r+2) - (2r+1))}{2.4.6.8 \dots (2r+2)}$$

$$= \frac{1.3.5 \dots (2r-1)(2r+2)}{2.4.6.8 \dots (2r+2)}$$

$$\frac{1.3.5 \dots (2r+1)}{2.4.6.8 \dots (2r+2)}$$

$$T_r = \frac{1.3.5 \dots (2r-1)}{2.4.6.8 \dots (2r)}$$

$$\frac{1.3.5 \dots (2r+1)}{2.4.6.8 \dots (2r+2)}$$

$$T_1 = \frac{1}{2} - \frac{1.3}{2.4}$$

$$T_2 = \frac{1.3}{2.4} - \frac{1.3.5}{2.4.6} \text{ and so.....on}$$

Finally it comes out to be $\frac{1}{2}$

20. **Ans (C)**

$$y = \begin{cases} 1, & x \in \mathbb{R} - (2n+1)\frac{\pi}{2}, n \in \mathbb{I} \\ e, & x \in (2n+1)\frac{\pi}{2}, n \in \mathbb{I} \end{cases}$$

Clearly two points of non differentiability.

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1274)**

$$\begin{aligned} & \frac{5}{6} + \frac{23}{12} + \frac{59}{20} + \frac{119}{30} + \frac{209}{42} + \dots n \text{ terms} \\ &= \left(1 - \frac{1}{6}\right) + \left(2 - \frac{1}{12}\right) + \left(3 - \frac{1}{20}\right) + \\ & \left(4 - \frac{1}{30}\right) + \left(5 - \frac{1}{42}\right) + \dots n \text{ terms} \\ &= \frac{n(n+1)}{2} - \\ & \left(\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \dots n \text{ terms}\right) \dots (1) \end{aligned}$$

Now T_r of $\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \dots$ is

$$\begin{aligned} T_r &= \frac{1}{r^2 + 3r + 2} = \frac{1}{(r+1)(r+2)} \\ &= \frac{1}{r+1} - \frac{1}{r+2} \\ \Rightarrow S_n &= \frac{1}{2} - \frac{1}{n+2} \end{aligned}$$

$\Rightarrow$ Sum of n terms of series,

$$\begin{aligned} S_n &= \frac{n(n+1)}{2} - \frac{1}{2} + \frac{1}{n+2} \\ \Rightarrow S_{50} &= \frac{50 \times 51}{2} - \frac{1}{2} + \frac{1}{52} \\ \Rightarrow \lambda &= 1275 - \frac{1}{2} + \frac{1}{52} \\ \Rightarrow [\lambda] &= 1274. \end{aligned}$$

2. **Ans (72)**

$$|A| = 2^4$$

$$|\text{adj}(2(AB + 16I))| = |2(AB + 2^4I)|^3$$

$$= 2^{12} ||A| + 2^4I|^3$$

$$= 2^{12} (|A| + 2^4)^{12} = 2^{72}$$

3. **Ans (5)**

These two curves symmetric about $y = x$ line and minimum distance will be distance between tangents whose slope is perpendicular

$$\therefore y = x^2 + x + 1 \Rightarrow \frac{dy}{dx} = 2x + 1 = 1$$

$$x = 0 \Rightarrow y = x^2 + x + 1 \Rightarrow y = 1$$

$$m = \frac{y - y_1}{x - x_1} \Rightarrow 1 = \frac{y - 1}{x - 0}$$

$$x = y - 1 \Rightarrow x - y + 1 = 0$$

$$\Rightarrow x = y^2 + y + 1$$

$$1 = 2y \cdot \frac{dy}{dx} + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{2y+1}$$

$$\Rightarrow 1 = \frac{1}{2y+1}$$

$$\Rightarrow 2y + 1 = 1 \Rightarrow y = 0$$

$$\text{then } x = 1 \Rightarrow 1 = \frac{y-0}{x-1}$$

$$x - 1 = y \Rightarrow x - y - 1 = 0$$

$$\text{Now } x - y - 1 = 0 \text{ and } x - y + 1 = 0$$

$$k = \frac{2}{\sqrt{2}} \Rightarrow k = \sqrt{2}$$

$$\Rightarrow \frac{5k}{\sqrt{2}} = \frac{5 \times \sqrt{2}}{\sqrt{2}} = 5$$

4. **Ans (90)**

Domain of $f(x)$ is $[-1, 1]$

$$f(x) = 60x^4 - 60x^3 + 60x^2 - 60x + 60 + \frac{1}{\sqrt{1-x^2}}$$

$$f'(x) = 60 [x^4 - x^3 + x^2 - x + 1] + \frac{1}{\sqrt{1-x^2}}$$

$f(x)$ is strictly increasing in $[-1, 1]$

$$a = f(-1) = -12 - 15 - 20 - 30 - 60 - \frac{\pi}{2}$$

$$= -137 - \frac{\pi}{2}$$

$$b = f(1) = 12 - 15 + 20 - 30 + 60 + \frac{\pi}{2} = 47 + \frac{\pi}{2}$$

$$|a + b| = \left| -137 - \frac{\pi}{2} + 47 + \frac{\pi}{2} \right| = 90$$

5. Ans (50)

$$\int_0^a f(x) dx = \frac{a^2}{2} + \frac{a}{2} \sin a + \frac{\pi}{2} \cos a$$

Using Leibnitz Rule

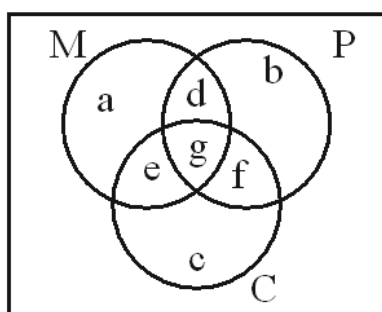
$$f(a) = a + \frac{a}{2} \cdot \cos a + \frac{\sin a}{2} - \frac{\pi}{2} \sin a$$

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + \frac{1}{2} - \frac{\pi}{2}$$

$$f\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

$$100 F\left(\frac{\pi}{2}\right) = 100 \times \frac{1}{2} = 50$$

6. Ans (2)



$$a + b + c + d + e + f + g = 120$$

$$d + e + f = 46$$

$$g = 11$$

$$\Rightarrow a + b + c = 63$$

$$\text{Also, } \frac{b}{c} = \frac{3}{5} \Rightarrow b = \frac{3c}{5}$$

$$\Rightarrow a + \frac{3c}{5} + c = 63$$

$$\Rightarrow a + \frac{8c}{5} = 63 \dots\dots (1)$$

from equation ... (1)

$$\Rightarrow c = 5k \quad (k \geq 3)$$

$$\Rightarrow a = 63 - 8k$$

$$\Rightarrow b = 3k$$

minimum value of a, b, c is 13

$$\Rightarrow 3k > 13 \Rightarrow (k \geq 5)$$

$$\text{Also } 5k > 13 \Rightarrow (k \geq 3)$$

$$63 - 8k > 13 \Rightarrow 8k < 50 \Rightarrow (k \leq 6)$$

only possible value of k are 5 and 6

$\Rightarrow$ 2 such combination are only possible.

7. Ans (99)

$$5000 - 4950 + 100 - 1 \quad C_{100-1} = {}^{149}C_{99} \text{ or } {}^{149}C_{50}$$

8. Ans (18)

$$\begin{aligned} & \int_3^6 \left(\sqrt{x+2\sqrt{3(x-3)}} + \sqrt{x-2\sqrt{3(x-3)}} \right) dx \\ &= \int_3^6 \left(\sqrt{(\sqrt{x-3})^2 + (\sqrt{3})^2 + 2\sqrt{3} \cdot \sqrt{x-3}} + \sqrt{(\sqrt{x-3})^2 + (\sqrt{3})^2 - 2\sqrt{3} \cdot \sqrt{x-3}} \right) dx \\ &= \int_3^6 \left(\sqrt{(\sqrt{3} + \sqrt{x-3})^2} + \sqrt{(\sqrt{3} - \sqrt{x-3})^2} \right) dx \\ &= \int_3^6 (\sqrt{3} + \sqrt{x-3} + \sqrt{3} - \sqrt{x-3}) dx \\ &= 2\sqrt{3} \int_3^6 dx = 6\sqrt{3} \end{aligned}$$

9. Ans (10)

$$\text{At } z = x + iy$$

$$C : (2iy - 6i)^2 + 8(2x - 4) = 0$$

$$\Rightarrow -4(y - 3)^2 = -16(x - 2)$$

$$\Rightarrow (y - 3)^2 = 4 \cdot 1 \cdot (x - 2)$$

A and B are ends of a double ordinate.

Equation of tangent at A(6, 7) is $x - 2y + 8 = 0$.

$$\Rightarrow T(-2, 3) \Rightarrow \text{Centroid of } \Delta TAB \text{ is } \left(\frac{10}{3}, 3 \right)$$

10. Ans (5)

For $x = 1$, $0 = f(y) + f\left(\frac{1}{y}\right)$

$$2f(x) = f(xy) + f\left(\frac{x}{y}\right) \quad \dots\dots(i)$$

Interchanging x and y we get

$$2f(y) = f(yx) + f\left(\frac{y}{x}\right) \quad \dots\dots(ii)$$

Adding equation (i) and (ii)

$$2f(x) + 2f(y) = 2f(xy) + f\left(\frac{x}{y}\right) + f\left(\frac{y}{x}\right)$$

But $f\left(\frac{x}{y}\right) + f\left(\frac{y}{x}\right) = 0$

So $f(x) + f(y) = f(xy)$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f\left(x \cdot \left(1 + \frac{h}{x}\right)\right) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x) + f\left(1 + \frac{h}{x}\right) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right) - f(1)}{\frac{h}{x} \cdot x}, \quad f'(x) = \frac{f'(1)}{x}
 \end{aligned}$$

$$f'(x) = \frac{1}{\ln 6 \cdot x}$$

Integrating with respect to 'x'

$$f(x) = \frac{1}{\ln 6} \cdot \ln|x| + C$$

$$f(1) = \frac{1}{\ln 6} \cdot 0 + C = 0 \Rightarrow C = 0$$

$$f(x) = \frac{\ln|x|}{\ln 6} = \frac{\ln x}{\ln 6} \text{ as } x > 0$$

$$\text{Now, } f(7776) = \frac{\ln(7776)}{\ln 6} = \frac{\ln(6^5)}{\ln 6} = 5$$